



Sensor that monitors firefighters' protective gear for safety



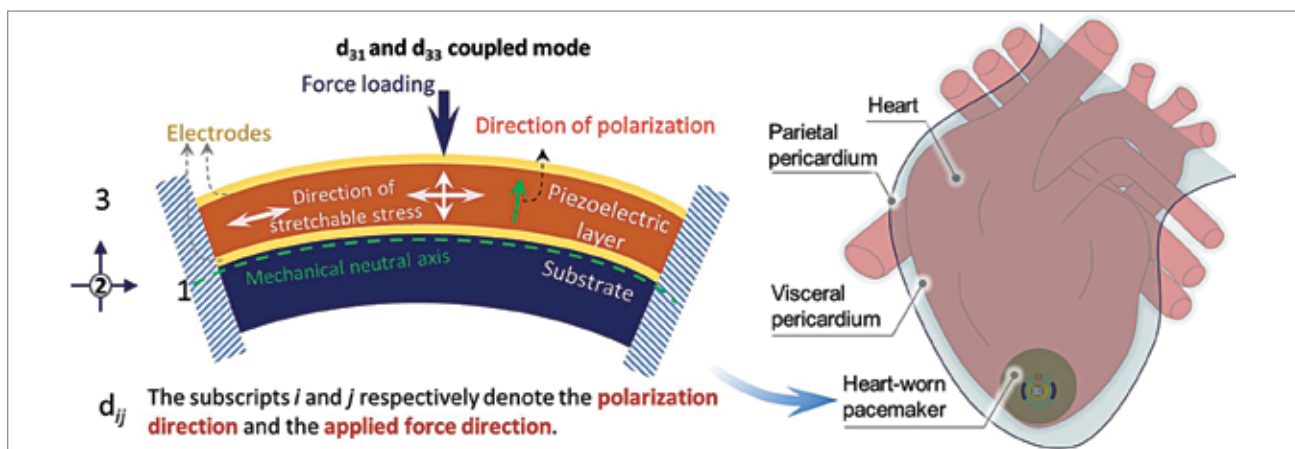
A sensor developed at the University of Alberta could help protect firefighters and others in hazardous work situations by detecting deterioration in their protective clothing (Credit: www.ualberta.ca)

A University of Alberta researcher is working with industry to develop a sensor that can detect the gradual breakdown in garments from exposure to heat, moisture, and ultraviolet (UV) light and reduce the risk of fatal injuries/burns caused by fire. Damage to the garments may not be visible to the naked eye before performance gets reduced considerably. Once fully developed, the sensor patch would provide a way to assess the garment without destructive testing. The sensor patch uses graphene to form conductive tracks on the patch's surface. When exposure to heat, moisture, or UV light exceeds a certain level, the graphene track is disrupted and loses its electrical conductivity. Firefighters would simply need to use a voltmeter to check the safety levels of their clothing on the sensor patch.

Batteryless cardiac pacemaker powered by heart's kinetic energy

Existing cardiac pacemakers are battery-powered devices that use leads to maintain the heartbeats. But batteries lead to infections and are complex to install. Researchers from the Shanghai Jiao Tong University have proposed a technique to harvest kinetic energy from the heart to power the lifesaving device. The energy is harvested by the buckling of the encapsulated structure of the pacemaker, creating buckled

piezoelectric energy. This method of power supply varies greatly from current pacemaker designs. The batteryless pacemaker is in the feasible verification phase. The research team is working hard to overcome some drawbacks such as long-term stability in vivo, implanting methods, and integration between the rigid pacing chip and the flexible energy harvesting unit.



The cardiac pacemaker of the future could be powered by the heart itself (Credit: <https://scitechdaily.com>)